

Linearized Drift-Pressure Model: Definitions, Equilibrium, and the Linear Stability Matrix

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Overview

This note collects the working derivation into a single, self-contained reference. Section 1 fixes notation and lists every constant/operator that appears, with its physical meaning and (where defined) its formula. Section 2 states the two governing equations. Section 3 derives the

axisymmetric, steady equilibrium. Section 4 introduces the perturbation ansatz and linearizes both equations term by term. Section 5 assembles the 2×2 linear operator (eigenvalue) problem, giving *both* the compact and the fully expanded operator entries. Section 8 collects every approximation and open question.

1 Notation, Operators, and Constants

1.1 Bracket and Laplacian operators

Generalized Poisson bracket. For two scalar fields $a(r, \theta)$, $b(r, \theta)$,

$$\{a, b\} \equiv a_x b_y - a_y b_x, \quad (1)$$

where subscripts denote partial derivatives ($a_x = \partial a / \partial x$, etc.). In polar coordinates this is equivalently

$$\{a, b\} = \frac{1}{r} (a_r b_\theta - a_\theta b_r). \quad (2)$$

Laplacian.

$$\Delta \equiv \nabla^2. \quad (3)$$

When axisymmetry is assumed (no θ -dependence in the background), the full Laplacian acting on a perturbation $\propto e^{im\theta + \gamma t}$ separates as

$$\Delta \rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}, \quad (4)$$

and we define the m -th harmonic radial Laplacian operator

$$\Delta_m \equiv \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) - \frac{m^2}{r^2}, \quad (5)$$

i.e. $\Delta(\hat{\varphi} e^{im\theta + \gamma t}) = \Delta_m \hat{\varphi} e^{im\theta + \gamma t}$, since $\partial_\theta^2 \rightarrow -m^2$ on this harmonic.

Note: for an equilibrium quantity $f_0(r)$ that has *no* θ -dependence, the relevant Laplacian is the $m = 0$ case, $\Delta f_0 = \frac{1}{r} \frac{d}{dr} \left(r \frac{df_0}{dr} \right) = f_0'' + \frac{f_0'}{r}$, *not* $\Delta_m f_0$ (which would add a spurious $-m^2/r^2$ term).

1.2 Field variables

Symbol	Meaning
φ	Normalized plasma potential, $\varphi = \phi / \phi_\star$, where $\phi_\star$ is the (central) amplitude of the applied wall potential.
P	Normalized pressure, $P = p / p_0$, where p_0 is the plasma pressure on axis at the trap midplane.
ρ	Mass density, $\rho = m_i n_0$ (ion mass $\times$ equilibrium number density).

1.3 Constants and coefficients

Symbol	Definition	Physical meaning
U	$U = \frac{p_{0i}}{e n \phi}$	FLR parameter (ion pressure / charge density $\times$ potential)
H	$H = \frac{\tau^2 e \phi}{c L T_e} \left(\frac{J_{i0}}{B} \right)_{\text{wall}} \frac{1}{\langle \rho B^{-2} \rangle}$	Plasma-wall coupling (line-tying) strength.
$\mathcal{K}$	$\mathcal{K} = 2 P_0 \tau^2 \frac{\langle g(z) \alpha(z) B^{-1} \rangle}{\langle \rho B^{-2} \rangle}$	Normalized average magnetic curvature.
J_{i0}	—	Equilibrium axial ion current.
τ	$\tau = \frac{B R^2}{c \phi}$	$E \times B$ drift period (BR^2 is const. on each field line).
L	—	Half-length of the trap.
$g(z)$	$\langle g \rangle = 1$	Axial pressure profile.
$\alpha(z)$	$\frac{1}{B(r, z)} = \frac{1}{B(0, z)} + \alpha(z) r^2$	Paraxial field expansion coefficient; obtainable from $\nabla \cdot \mathbf{B}$
T_e	—	Electron temperature.
φ_{fe}	—	Floating potential (flat in radius at constant T_e).
φ_{wall}	—	End-wall bias (applied externally).
φ_w	$\varphi_w = \varphi_{fe} + \varphi_{\text{wall}}$	Total effective wall potential.
$\nu_{\perp}^P$	—	Collisional transverse (cross-field) diffusion rate acting on P .
$\nu_{\parallel}^P$	—	Axial diffusion rate acting on P .
$\nu_{\perp}, \nu_{\parallel}$	—	Transverse/axial diffusion rates acting on φ (not separate).
Q_P	$Q_P \approx Q_P(r)$	Power source term in the pressure equation.
S_p	$S_p \approx (\nu_{\perp}^P - \nu_{\perp}) \Delta P - (\nu_{\parallel}^P - \nu_{\parallel}) P + Q_p$	Effects of external heating, axial outflow, and transverse t

Open notational issue: it is not fully specified whether φ_w is normalized with respect to ϕ in the same way as φ , i.e. whether $\varphi_w \equiv \phi_w / \phi_{\star}$ consistently.

2 Governing Equations

The starting point is a pair of coupled evolution equations for the normalized potential φ and normalized pressure P :

$$\partial_t \Delta \varphi + \{\varphi, \Delta \varphi\} - U \nabla \cdot \{\nabla \varphi, P\} = H(\varphi - \varphi_w) + \mathcal{K}\{P, r^2\} + \nu_{\perp} \Delta^2 \varphi - \nu_{\parallel} \Delta \varphi - U \Delta S_p, \quad (6)$$

$$\partial_t P + \{\varphi, P\} = \nu_{\perp}^P \Delta P - \nu_{\parallel}^P P + Q_P. \quad (7)$$

Equation (6) is the vorticity (potential) equation; Equation (7) is the pressure equation.

3 Equilibrium

Assumption E1 (axisymmetric, steady equilibrium). The background state has no θ -dependence and is time-independent: $\partial_t(\cdot)_0 = 0$ and $\partial_\theta(\cdot)_0 = 0$.

Under E1, any bracket of two equilibrium quantities vanishes since both fields are functions of r only, so every θ -derivative is zero. Equation (6) therefore collapses to

$$0 = H(\varphi_0 - \varphi_w) + \nu_\perp \Delta^2 \varphi_0 - \nu_\parallel \Delta \varphi_0 - U \Delta S_{P,0}. \quad (8)$$

Assumption E2. The diffusive terms $\nu_\perp \Delta^2 \varphi_0$ and $\nu_\parallel \Delta \varphi_0$ are small compared to $H(\varphi_0 - \varphi_w)$.

Assumption E3. The axial and transverse diffusion rates for φ and P coincide: $\nu_\perp \approx \nu_\perp^P$ and $\nu_\parallel \approx \nu_\parallel^P$.

With E2 and E3 the equilibrium relation reduces to

$$H(\varphi_0 - \varphi_w) - U \Delta Q_p = 0 \implies \boxed{\varphi_0 \approx \varphi_w}. \quad (9)$$

The equilibrium condition for the *pressure* equation (Eq. (7) with E1 and Assumption L3 below) is

$$\nu_\parallel^P P_0 = Q_p \quad (\text{equilibrium pressure balance}). \quad (10)$$

3.1 Equilibrium rotation frequency

Since $\varphi_0 = \varphi_0(r)$ the equilibrium $E \times B$ flow is purely azimuthal with frequency

$$\Omega_0(r) \equiv \frac{1}{r} \frac{\partial \varphi_0}{\partial r} \propto \frac{1}{r} \frac{\partial \varphi_w}{\partial r}, \quad (11)$$

where the second equality uses Eq. (9).

4 Perturbation Ansatz and Linearization

Assumption L1 (normal-mode ansatz).

$$\varphi(r, \theta, t) = \varphi_0(r) + \hat{\varphi}(r) e^{im\theta + \gamma t}, \quad (12)$$

$$P(r, \theta, t) = P_0(r) + \hat{p}(r) e^{im\theta + \gamma t}. \quad (13)$$

Here $m \in \mathbb{Z}$ is the azimuthal mode number and $\gamma \in \mathbb{C}$ is the growth rate.

Assumption L2 (linearization). Only first-order terms in the perturbation amplitudes $(\hat{\varphi}, \hat{p})$ are retained; products of two perturbed quantities are dropped.

4.1 Linearized pressure equation

Starting from Eq. (7) and applying L1–L2 with Assumption L3 below.

Assumption L3. The cross-field diffusion term $\nu_{\perp}^P \Delta P$ is dropped in the linearized pressure balance.

The bracket splits into:

$$\{\varphi_0, \delta P\} = \frac{1}{r} \frac{\partial \varphi_0}{\partial r} \frac{\partial(\delta P)}{\partial \theta} = im \Omega_0(r) \hat{p} e^{im\theta + \gamma t}, \quad (14)$$

$$\{\delta \varphi, P_0\} = -\frac{1}{r} \frac{\partial(\delta \varphi)}{\partial \theta} \frac{\partial P_0}{\partial r} = -\frac{im}{r} P_0'(r) \hat{\varphi} e^{im\theta + \gamma t}. \quad (15)$$

After separating orders in $e^{im\theta + \gamma t}$:

Order-zero (equilibrium): Eq. (10) is recovered: $\nu_{\parallel}^P P_0 = Q_p$.

Order-one (perturbed):

$$\gamma \hat{p} + im \Omega_0 \hat{p} - \frac{im}{r} P_0'(r) \hat{\varphi} = -\nu_{\parallel}^P \hat{p}, \quad (16)$$

which rearranges to

$$\gamma \hat{p} = \frac{im}{r} P_0'(r) \hat{\varphi} - (\nu_{\parallel}^P + im \Omega_0) \hat{p}. \quad (17)$$

4.2 Linearized vorticity equation

Now linearize Eq. (6).

Assumption L4. The hyper-diffusive term $\nu_{\perp} \Delta^2 \varphi$ and the source-gradient term $U \Delta S_P$ are dropped in the linear stability analysis.

Assumption L5. Products of two perturbed quantities (“frightening” terms) are discarded (part of strict linearization L2).

Term 1: $\partial_t \Delta \varphi$

$$\partial_t \Delta \varphi \longrightarrow \gamma \Delta_m \hat{\varphi} e^{im\theta + \gamma t}. \quad (18)$$

Term 2: $\{\varphi, \Delta\varphi\}$

Expand $\Delta\varphi = \Delta\varphi_0 + \Delta_m\delta\varphi$. The bracket splits into three pieces; the equilibrium–equilibrium piece vanishes by E1:

$$\{\varphi_0, \Delta\varphi_0\} = 0, \quad (19)$$

$$\{\delta\varphi, \Delta\varphi_0\} = -\frac{im}{r}\hat{\varphi}e^{im\theta+\gamma t}(\Delta\varphi_0)', \quad (20)$$

$$\{\varphi_0, \Delta\delta\varphi\} = im\Omega_0\Delta_m\hat{\varphi}e^{im\theta+\gamma t}. \quad (21)$$

Here primes denote d/dr , and $(\Delta\varphi_0)'$ is the radial derivative of $\Delta\varphi_0 = \varphi_0'' + \varphi_0'/r$ (the *full* Laplacian of the equilibrium potential, with no m^2/r^2 correction).

Term 3: $U\nabla \cdot \{\nabla\varphi, P\}$

Using $\nabla A = \hat{r}\partial_r A + \hat{\theta}(1/r)\partial_\theta A$ and $\nabla \cdot \mathbf{A} = (1/r)\partial_r(rA_r) + (1/r)\partial_\theta A_\theta$, the two linearized contributions are

$$\nabla \cdot \{\nabla\varphi_0, \delta P\} = \frac{1}{r}\frac{\partial}{\partial r}[im\hat{p}\varphi_0'']e^{im\theta+\gamma t}, \quad (22)$$

$$\nabla \cdot \{\nabla\delta\varphi, P_0\} = \frac{1}{r}\frac{\partial}{\partial r}[-imP_0'\hat{\varphi}']e^{im\theta+\gamma t} + \frac{im^3}{r^3}P_0'\hat{\varphi}e^{im\theta+\gamma t}, \quad (23)$$

which combine to

$$\nabla \cdot \{\nabla\varphi, P\}\Big|_{\text{linear}} = \left[\frac{1}{r}\frac{\partial}{\partial r}(im\hat{p}\varphi_0'' - imP_0'\hat{\varphi}') + \frac{im^3}{r^3}P_0'\hat{\varphi}\right]e^{im\theta+\gamma t}. \quad (24)$$

Term 4: $\mathcal{K}\{P, r^2\}$

Since r^2 has no θ -dependence, only $\{\delta P, r^2\}$ survives:

$$\mathcal{K}\{P, r^2\}\Big|_{\text{linear}} = -2im\mathcal{K}\hat{p}e^{im\theta+\gamma t}. \quad (25)$$

Assembled linearized vorticity equation

Collecting all terms, using $\varphi_0 \approx \varphi_w$ (so $H(\varphi_0 - \varphi_w) \rightarrow 0$ for the equilibrium and $H\hat{\varphi}$ for the perturbation), and dividing by $e^{im\theta+\gamma t}$:

$$\begin{aligned} \gamma\Delta_m\hat{\varphi} = & -im\Omega_0\Delta_m\hat{\varphi} + \frac{im}{r}(\Delta\varphi_0)'\hat{\varphi} - \frac{imU}{r}\frac{\partial}{\partial r}(P_0'\hat{\varphi}') + \frac{im^3U}{r^3}P_0'\hat{\varphi} \\ & + H\hat{\varphi} - \nu_{\parallel}\Delta_m\hat{\varphi} + \frac{imU}{r}\frac{\partial}{\partial r}(\varphi_0''\hat{p}) - 2im\mathcal{K}\hat{p}. \end{aligned} \quad (26)$$

Here $(\Delta\varphi_0)' = \varphi_0''' + \varphi_0''/r - \varphi_0'/r^2$ is the radial derivative of the equilibrium Laplacian $\Delta\varphi_0 = \varphi_0'' + \varphi_0'/r$. This term represents the shear in equilibrium vorticity and is kept as an active contribution.

5 The Linear Eigenvalue System

Writing the system as a generalized eigenvalue problem with eigenvalue γ :

$$\gamma \begin{pmatrix} \Delta_m & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix} = \begin{pmatrix} \mathcal{L}_{11} & \mathcal{L}_{12} \\ \mathcal{L}_{21} & \mathcal{L}_{22} \end{pmatrix} \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix}. \quad (27)$$

5.1 Compact (as-derived) operator entries

The entries read directly from Eqs. (17) and (26), using Δ_m^{-1} to convert any $\hat{\varphi}$ back to $\Delta_m \hat{\varphi}$:

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' - \frac{imU}{r} \frac{\partial}{\partial r} \left(P_0' \frac{\partial}{\partial r} \right) + \frac{im^3U}{r^3} P_0' + H - \nu_{\parallel} \Delta_m, \quad (28)$$

$$\mathcal{L}_{12} = \frac{imU}{r} \frac{\partial}{\partial r} (\varphi_0'' \cdot) - 2im\mathcal{K}, \quad (29)$$

$$\mathcal{L}_{21} = \frac{im}{r} P_0'(r) \quad (30)$$

$$\mathcal{L}_{22} = -\nu_{\parallel}^P - im\Omega_0. \quad (31)$$

5.2 Expanded operator entries (product rule applied explicitly)

$\mathcal{L}_{11}$ expanded

Apply the product rule to the FLR term:

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' - \frac{imU}{r} \frac{\partial}{\partial r} \left(P_0' \frac{\partial}{\partial r} \right) + \frac{im^3U}{r^3} P_0' + H - \nu_{\parallel} \Delta_m \quad (32)$$

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' - \frac{imU}{r} (P_0'' \frac{\partial}{\partial r} + P_0' \frac{\partial^2}{\partial r^2}) + \frac{im^3U}{r^3} P_0' + H - \nu_{\parallel} \Delta_m \quad (33)$$

where $(\Delta\varphi_0)' = \varphi_0''' + \varphi_0''/r - \varphi_0'/r^2$.

$\mathcal{L}_{12}$ expanded

$\mathcal{L}_{12}$ acts on $\hat{p}$:

$$\begin{aligned} \mathcal{L}_{12} \hat{p} &= \frac{imU}{r} \frac{d}{dr} (\varphi_0'' \hat{p}) - 2im\mathcal{K} \hat{p} \\ &= \frac{imU}{r} (\varphi_0''' \hat{p} + \varphi_0'' \hat{p}') - 2im\mathcal{K} \hat{p}. \end{aligned} \quad (34)$$

$\mathcal{L}_{21}$ and $\mathcal{L}_{22}$

Both are already pure multiplication operators (no derivatives):

$$\mathcal{L}_{21} \hat{\varphi} = \frac{im}{r} P_0'(r) \cdot \hat{\varphi}, \quad (35)$$

$$\mathcal{L}_{22} \hat{p} = (-\nu_{\parallel}^P - im\Omega_0(r)) \cdot \hat{p}. \quad (36)$$

6 Boundary Conditions

We impose Dirichlet boundary conditions at the edge: $\varphi(r = R) = 0$. Physically, this corresponds to the tungsten limiter surrounding the magnetic mirror.

7 Numerical Implementation

Spacing between radial points at which we evaluate $\hat{\varphi}$ and $\hat{P}$ is assumed to be some small value dr . The $\frac{1}{r}$ and $\frac{1}{r^2}$ terms in the PDE are undefined at $r = 0$, so we choose the zeroth entry in each of $\hat{\varphi}$ and $\hat{P}$ to correspond to the radial position $\frac{dr}{2}$. Furthermore, we assume φ and P are discretized to N radial positions.

7.1 Finite Difference Stencils

Grid points are $r_i = (i + \frac{1}{2}) dr$ for $i = 0, 1, \dots, N - 1$. All stencils are second-order accurate, $O(dr^2)$.

First derivative

Interior ($1 \leq i \leq N - 2$):

$$f'_i \approx \frac{-f_{i-1} + f_{i+1}}{2 dr}. \quad (37)$$

Left boundary ($i = 0$, forward):

$$f'_0 \approx \frac{-3f_0 + 4f_1 - f_2}{2 dr}. \quad (38)$$

Right boundary ($i = N - 1$, backward):

$$f'_{N-1} \approx \frac{f_{N-3} - 4f_{N-2} + 3f_{N-1}}{2 dr}. \quad (39)$$

Second derivative

Interior ($1 \leq i \leq N - 2$):

$$f''_i \approx \frac{f_{i-1} - 2f_i + f_{i+1}}{dr^2}. \quad (40)$$

Left boundary ($i = 0$, forward):

$$f''_0 \approx \frac{2f_0 - 5f_1 + 4f_2 - f_3}{dr^2}. \quad (41)$$

Right boundary ($i = N - 1$, backward):

$$f''_{N-1} \approx \frac{-f_{N-4} + 4f_{N-3} - 5f_{N-2} + 2f_{N-1}}{dr^2}. \quad (42)$$

Third derivative

Interior ($2 \leq i \leq N-3$):

$$f_i''' \approx \frac{-f_{i-2} + 2f_{i-1} - 2f_{i+1} + f_{i+2}}{2dr^3}. \quad (43)$$

Near-left boundary ($i = 1$):

$$f_1''' \approx \frac{-3f_0 + 10f_1 - 12f_2 + 6f_3 - f_4}{2dr^3}. \quad (44)$$

Left boundary ($i = 0$, forward):

$$f_0''' \approx \frac{-5f_0 + 18f_1 - 24f_2 + 14f_3 - 3f_4}{2dr^3}. \quad (45)$$

Near-right boundary ($i = N-2$):

$$f_{N-2}''' \approx \frac{f_{N-5} - 6f_{N-4} + 12f_{N-3} - 10f_{N-2} + 3f_{N-1}}{2dr^3}. \quad (46)$$

Right boundary ($i = N-1$, backward):

$$f_{N-1}''' \approx \frac{3f_{N-5} - 14f_{N-4} + 24f_{N-3} - 18f_{N-2} + 5f_{N-1}}{2dr^3}. \quad (47)$$

7.2 Imposing Boundary Conditions Numerically

Recall that the eigenvalue problem we seek to solve is:

$$\gamma A \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix} = B \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix} \quad (48)$$

with $A = \begin{pmatrix} \Delta_m & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} \mathcal{L}_{11} & \mathcal{L}_{12} \\ \mathcal{L}_{21} & \mathcal{L}_{22} \end{pmatrix}$. We impose the boundary condition $\varphi(r = R) = 0$ by setting the Nth row of A to zeros and the Nth row of B to all zeros except for a single 1 in the Nth column.

8 Summary of Approximations and Open Questions

8.1 Equilibrium approximations

- **(E1)** Axisymmetric, steady background: $\partial_\theta(\cdot)_0 = 0$, $\partial_t(\cdot)_0 = 0$. Kills every equilibrium-equilibrium bracket.
- **(E2)** Diffusion subdominant at equilibrium: $\nu_\perp \Delta^2 \varphi_0$ and $\nu_\parallel \Delta \varphi_0$ dropped relative to $H(\varphi_0 - \varphi_w)$, giving $\varphi_0 \approx \varphi_w$.
- **(E3)** $\nu_\parallel \approx \nu_\parallel^P$ and $\nu_\perp \approx \nu_\perp^P$ (equal diffusion rates for φ and P).
- *Open question:* Is $\nu_\parallel \approx \nu_\parallel^P$ valid?

- *Open question:* Is $\nu_{\perp} \ll \nu_{\parallel}$ (transverse diffusion negligible)?
- *Open question:* Is φ_w normalized consistently with φ (both divided by $\phi_{\star}$)?
- *Open question:* Can the axial pressure profile be approximated as $g(z) = 1$, given $\langle g \rangle = 1$ by definition?

8.2 Perturbation ansatz and linearization

- **(L1)** Single-harmonic normal-mode ansatz.
- **(L2)** Strict linearization: only terms first-order in $(\hat{\varphi}, \hat{p})$ kept.
- **(L3)** $\nu_{\perp}^P \Delta P \rightarrow 0$ in the linearized pressure balance.
- **(L4)** $\nu_{\perp} \Delta^2 \varphi$ and $U \Delta S_P$ dropped in the vorticity equation.
- **(L5)** No second-order (“frightening”) terms.

8.3 Undefined or implicitly defined quantities

- $\nu_{\perp}$ and $\nu_{\parallel}$ (diffusion rates in the vorticity equation) are used but never explicitly defined in the source material.
- $\alpha(z)$ is defined implicitly via $1/B(r, z) = 1/B(0, z) + \alpha(z)r^2$ and must be computed from $\nabla \cdot \mathbf{B} = 0$ with the known on-axis field $B(0, z)$.